HIKET — Bayesian Calibration of Forest SOC Models

Methods and pipeline documentation (transient pre-initialisation)

Lorenzo Menichetti

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Contents

1 Overview

This document describes the Bayesian calibration framework underlying the HIKET intercomparison of soil organic carbon (SOC) decomposition models for Finnish forest ecosystems. The pipeline is implemented in a model-agnostic engine (`calibration_engine_transient.R`) and five model-specific orchestration scripts.

1.1 Research question

Divergent projections of the Finnish forest carbon sink saturation point emerge from existing SOC models. Are these differences a property of model *structure*, or artefacts of *calibration*? HIKET addresses this by calibrating five models against the same data, with identical likelihood, data filtering, and observation-error assumptions, so that residual differences are attributable to structure rather than to parametric freedom.

1.2 Models

Five models span a complexity ladder crossed with two climate-integration choices:

Model	Pools	Free params	Climate ξ	Notes
SP1	1	6	single triplet, Yasso07	Pure R
TP2	2	8	single triplet, Yasso07	ICBM topology, pure R

Model	Pools	Free params	Climate ξ	Notes
Yasso07	5	18	single triplet	Fortran core
Yasso15	5	28	pool-group-specific	Fortran core (yasso15.so)
Yasso20	5	28	pool-group-specific, monthly T	Same Fortran as Yasso15

SP1 $\to$ TP2 $\to$ Yasso07 isolates the effect of adding pools ($1 \to 2 \to 5$). Yasso07 $\to$ Yasso15 isolates the effect of pool-group-specific climate response (single $\xi \to$ three ξ). Yasso15 $\to$ Yasso20 isolates the effect of intra-annual temperature integration (Gaussian quarterly approximation $\to$ direct monthly average). All other modelling choices — AWEN fractionation, decomposition rates, observation error, likelihood form, prior structure, initialisation strategy — are held identical wherever possible.

SP1 and TP2 share the Yasso07 ξ functional form, with independently calibrated $(\beta_1, \beta_2, \gamma)$. This guarantees that the climate-response comparison across the complexity ladder is not confounded by approximation-scheme differences.

1.3 Pipeline

Script	Purpose
`Data_work.R`	Data preparation (separately documented)
`run_{Model}_transient_calibration.R`	MCMC calibration; produces posterior samples
`run_{Model}_transient_predictive.R`	Posterior predictive trajectories, residuals
`run_residual_analysis.R`	Random-forest residual analysis (model-agnostic)
`run_multimodel_comparison.R`	Cross-model metrics and overlay plots

This document covers the calibration stage only. Data preparation is documented in `Data_work.R` itself.

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2 Models

All five models follow first-order kinetics: each pool decomposes at a rate proportional to its size, modulated by a climate modifier ξ. The litter input J is partitioned by chemical composition (AWEN: acid-, water-, ethanol-, and non-soluble fractions), and the partitioning is fixed from the input data.

2.1 SP1 (single pool)

All litter inputs — NWL (non-woody), FWL (fine woody), CWL (coarse woody), summed across AWEN fractions — enter a single pool C which decomposes by first-order kinetics:

$$\frac{dC}{dt} = J(t) - \alpha\,\xi(t)\,C(t).$$

The annual step has an exact closed-form solution (piecewise-constant J and ξ within each year):

$$C(t+1) = C_{\text{ss}}(t) + \big(C(t) - C_{\text{ss}}(t)\big)\, e^{-\alpha\xi(t)}, \qquad C_{\text{ss}}(t) = \frac{J(t)}{\alpha\,\xi(t)}.$$

Pure R implementation; no Fortran.

2.2 TP2 (two pools, ICBM topology)

Two pools, A (active) and H (humus). All litter enters A; H receives transferred mass from A only:

$$\frac{dA}{dt} = J(t) - \alpha_A \xi(t) A(t), \qquad \frac{dH}{dt} = p_H \alpha_A \xi(t) A(t) - \alpha_H \xi(t) H(t).$$

This is the ICBM topology of Andrén & Kätterer (1997), with the modification that ξ multiplies *both* pool decay rates (ICBM uses separate climate factors). Parameter naming follows Yasso07 (α_A, α_H, p_H), making the TP2 parameter set a *strict subset* of Yasso07 and clarifying the structural comparison.

The annual step is solved exactly, not Euler-approximated. With $k_A = \alpha_A\xi$, $k_H = \alpha_H\xi$, $A_{\text{ss}} = J/k_A$, and $\Delta_A = A(t) - A_{\text{ss}}$:

$$A(t+1) = A_{\text{ss}} + \Delta_A\, e^{-k_A}$$

$$H(t+1) = H(t)\, e^{-k_H} + \frac{p_H J}{k_H}(1 - e^{-k_H}) + \frac{p_H k_A \Delta_A}{k_H - k_A}(e^{-k_A} - e^{-k_H}).$$

The degenerate case $k_A = k_H$ is handled by the L'Hôpital limit ($p_H k_A \Delta_A e^{-k_A}$ for the last term). The analytical steady states are $A^* = J/(\alpha_A \xi)$ and $H^* = p_H J/(\alpha_H \xi)$; ξ cancels in the ratio H^*/A^*. Pure R; no Fortran.

2.3 Yasso07

Five pools (Tuomi et al. 2009, 2011): A (acid-soluble), W (water-soluble), E (ethanol-soluble), N (non-soluble / lignin), H (humus). Litter inputs enter A, W, E, N proportionally to their AWEN fractions; transfers between AWEN pools are controlled by 12 fractions p_{ij}; all four AWEN pools contribute to H at a humification rate p_H; H decomposes irreversibly (α_H).

The dynamics are written as a linear system $dC/dt = AC + b(t)$, with the matrix A assembled from the decomposition rates and transfer fractions and the input vector b from AWEN-fractionated litter. The Fortran core solves each annual step analytically through the matrix exponential $C(t+1) = A^{-1}(e^{A\Delta t}(AC(t)+b) - b)$, with two well-tested fallback paths (near-zero diagonal: pure accumulation; near-singular matrix: Euler step). FWL and CWL inputs are decomposed separately as parallel cohorts; the per-cohort state has 15 elements (5 pools $\times$ 3 size classes).

The climate modifier is a four-point Gaussian approximation of the intra-annual temperature cycle:

$$\xi(\bar{T}, T_{\text{amp}}, P) = \frac{1}{4}\sum_{k=1}^{4} \exp\!\big(\beta_1 T_k + \beta_2 T_k^2\big) \times \big(1 - \exp(\gamma P/1000)\big),$$

where T_k are four quarterly temperatures derived from annual mean $\bar{T}$ and seasonal amplitude T_{amp}. A single $(\beta_1, \beta_2, \gamma)$ triplet applies to all decomposition rates.

A woody-size modifier scales decomposition for FWL/CWL cohorts:

$$h_S(d) = \min\Big\{1,\ (1 + \delta_1 d + \delta_2 d^2)^{-|r|}\Big\},$$

where d is the cohort diameter (cm). Fixed diameters $d_{\text{fwl}} = 2$ cm and $d_{\text{cwl}} = 15$ cm are used throughout the pipeline.

2.4 Yasso15

Identical pool topology, AWEN partitioning, and matrix-exponential machinery as Yasso07. The structural change is *pool-group-specific* climate response: three triplets $(\beta_1, \beta_2, \gamma)$, $(\beta_{N1}, \beta_{N2}, \gamma_N)$, $(\beta_{H1}, \beta_{H2}, \gamma_H)$ act on the AWE, N, and H pool groups respectively. The functional form of each ξ matches Yasso07 (four-point Gaussian temperature approximation, exponential-with-saturation precipitation response).

Leaching parameters $w_1, \ldots, w_5$ are present in the parameter vector (positions 31–35) but inactive in the calibration: the leaching scalar argument to the Fortran is held at zero throughout. Yasso15 uses a 35-element parameter vector.

2.5 Yasso20

Same Fortran core as Yasso15 (`yasso15.so` is reused) and the same 35-parameter vector. The single structural change is the intra-annual temperature integration: instead of the four-point Gaussian approximation, the temperature modifier is averaged over all 12 monthly temperatures directly,

$$\xi_T^{(g)} = \frac{1}{12} \sum_{m=1}^{12} \exp\!\big(\beta_1^{(g)} T_m + \beta_2^{(g)} T_m^2\big), \qquad g \in \{\text{AWE}, N, H\},$$

with the same exponential precipitation factor applied to the annual total precipitation. This is the only computational difference between Yasso15 and Yasso20. Both share the same R wrapper functions for the matrix-exponential solver; only the ξ computation differs.

Yasso20 therefore inherits the same parameter structure as Yasso15 in this calibration (all 12 inter-pool flow fractions free; see §3.3).

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3 Parameter specification

3.1 Free, fixed, and derived parameters

For each model the parameter set splits into three groups:

- **Free**: sampled by MCMC. Includes nuisance parameters σ_{init}, σ_{input}.
- **Fixed**: not sampled, held at published or structurally-motivated values. Includes decomposition rates and humus parameters for the Yasso family.

- **Derived**: computed deterministically from other parameters at assembly time. *None* in the current pipeline.

For each model the parameters are listed below with their physical meaning and transform. Transforms map a constrained physical parameter to an unconstrained sampling space, with a corresponding log-Jacobian contribution to the prior density. Three transform types are used:

- `log`: enforces positivity; $\theta \in (0, \infty) \leftrightarrow x \in \mathbb{R}$, $x = \log \theta$.
- `logit`: enforces $(0, 1)$ bounds; $\theta \in (0, 1) \leftrightarrow x \in \mathbb{R}$, $x = \log(\theta/(1 - \theta))$.
- `unconstrained`: identity ($\theta = x$); used where the parameter has no natural constraint and the prior alone shapes the support.
- `stick_break`: a K-element simplex constrained to sum $\leq B$, used for transfer fractions; described separately in §3.4.

3.2 SP1 (6 free, 0 fixed)

Parameter	Meaning	Transform
α	Decomposition rate (yr^{-1})	log
β_1	Temperature response, linear	log
β_2	Temperature response, quadratic	unconstrained
γ	Precipitation response	unconstrained
σ_{init}	1917 litter scaling (initial state)	log
σ_{input}	Global litter scaling (contemporary)	log

No fixed parameters. SP1 has no published global calibration to anchor the kinetic rate, so all parameters are learned from the Finnish data.

3.3 TP2 (8 free, 0 fixed)

Parameter	Meaning	Transform
α_A	A-pool decomposition rate (yr^{-1})	log
α_H	H-pool decomposition rate (yr^{-1})	log
p_H	Humification fraction (A $\rightarrow$ H)	logit
β_1, β_2, γ	Climate response (Yasso07 form)	log, unc., unc.
$\sigma_{\mathrm{init}}, \sigma_{\mathrm{input}}$	Litter scalings	log, log

3.4 Yasso07 (18 free, 6 fixed)

Fixed (6): $\alpha_A, \alpha_W, \alpha_E, \alpha_N$ (AWEN decomposition rates), p_H (humification fraction), α_H (humus decomposition rate). Published Tuomi et al. defaults.

Free (18):

Parameter	Count	Transform
12 inter-pool transfer fractions p_{ij}, 4 columns of 3	12	stick_break
β_1, β_2, γ	3	log, unc., unc.

Parameter	Count	Transform
δ_1, δ_2, r (size modifier)	3	unc., log, log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	2	log, log

The size modifier transforms: $\delta_2 > 0$ is enforced because the Fortran returns NaN for $\delta_2 < 0$ when $d \gtrsim 0.4$ cm. The log transform on r removes the sign-redundancy of $|r|$ in the size formula (r is stored as positive in HIKET; Yasso07 reference defaults store it negative).

3.5 Yasso15 (28 free, 11 fixed)

Fixed (11): $\alpha_A, \alpha_W, \alpha_E, \alpha_N$, p_H, α_H, w_1, w_2, w_3, w_4, w_5 (leaching weights, inactive at leac = 0). Published Yasso15 defaults.

Free (28):

Parameter	Count	Transform
12 inter-pool transfer fractions p_{ij}	12	stick_break
AWE climate: β_1, β_2, γ	3	log, unc., unc.
N climate: $\beta_{N1}, \beta_{N2}, \gamma_N$	3	log, unc., unc.
H climate: $\beta_{H1}, \beta_{H2}, \gamma_H$	3	log, unc., unc.
Size modifier: δ_1, δ_2, r	3	unc., log, log
$\sigma_{\text{init}}, \sigma_{\text{input}}$	2	log, log

3.6 Yasso20 (28 free, 11 fixed)

Identical fixed/free structure to Yasso15. The same 35-parameter vector is used; only the ξ computation method differs at runtime.

Uniform 12-fraction structure across the Yasso family. Earlier Yasso20 calibrations (Viskari 2022) fixed six transfer fractions at zero (empirical structural zeros from global-dataset MCMC) and derived three (p_{AW}, p_{NA}, p_{EA}) algebraically from p_H and p_{EW}. HIKET *does not* impose these constraints. All 12 inter-pool fractions are sampled in all three Yasso models. The reasoning is twofold:

1. The Viskari zeros are properties of a warm-site global calibration dataset, not general biological laws. Finnish boreal data, where the recycling-loop dynamics dominate decomposition in cold sites, should be free to inform the full pool-flow structure.
2. If Yasso20 had fewer free fractions than Yasso07 and Yasso15, any calibration-quality difference between the models would be uninterpretable: structural (climate-integration) versus parametric (missing degrees of freedom).

Prior centres for the previously-zeroed fractions use the published Yasso15 defaults (which are nonzero MAP estimates from the same global calibration); see §4.

3.7 Stick-breaking parameterisation for transfer fractions

Each column j of the AWEN transfer matrix has three outflow fractions p_{ij} ($i \in \{A, W, E, N\}\ \{j\}$) that must satisfy $\sum_i p_{ij} + p_H \leq 1$ (mass conservation). The simplex constraint is enforced via a stick-breaking transform with budget $B = 1 - p_H$. Two unconstrained real-valued parameters z_1, z_2 map to three fractions (p_1, p_2, p_3) summing to $\leq B$ by

$$p_1 = B \cdot \sigma(z_1), \qquad p_2 = (B - p_1) \cdot \sigma(z_2), \qquad p_3 = B - p_1 - p_2,$$

where σ is the logistic function. The associated log-Jacobian is added to the prior density. Stick-breaking is preferred to logit on each fraction because (i) it guarantees the simplex constraint exactly, (ii) the Jacobian is well-conditioned, and (iii) the parameters are exchangeable in the sense that a swap of column order produces a mathematically equivalent posterior.

In Yasso07, Yasso15, and Yasso20, all four AWEN columns (A, W, E, N) use stick-breaking with the same budget $B = 1 - p_H$. Each column contributes 2 free parameters (the third is determined), for $4 \times 2 + 4 = 12$ fractions in total.

3.8 Cross-model parameter overlap

The following table summarises which parameters appear in which models:

Parameter	SP1	TP2	Yasso07	Yasso15	Yasso20
α (single rate)	free	—	—	—	—
α_A, α_H	—	free	fixed	fixed	fixed
$\alpha_W, \alpha_E, \alpha_N$	—	—	fixed	fixed	fixed
p_H	—	free	fixed	fixed	fixed
12 transfer fractions p_{ij}	—	—	free	free	free
β_1, β_2, γ	free	free	free	free (AWE)	free (AWE)
$\beta_{N1}, \beta_{N2}, \gamma_N$	—	—	—	free	free
$\beta_{H1}, \beta_{H2}, \gamma_H$	—	—	—	free	free
δ_1, δ_2, r	—	—	free	free	free
$w_1, \ldots, w_5$	—	—	—	fixed (=0)	fixed (=0)
$\sigma_{\text{init}}, \sigma_{\text{input}}$	free	free	free	free	free
Total free	**6**	**8**	**18**	**28**	**28**

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4 Priors

All priors are normal in the unconstrained (transformed) space, centred at the published model defaults. The prior density factorises across parameters; the joint prior is

$$p(x) = \prod_i \mathcal{N}(x_i\,;\, x_i^{\text{ctr}}, \sigma_{\text{ppm},i}) + \log J(x),$$

where x^{ctr} is the unconstrained-space image of the physical-space prior centre, σ_{ppm} is the unconstrained-space prior standard deviation, and $\log J$ is the log-Jacobian of the parameter transforms.

4.1 Sources

Prior centres and widths are derived from the most authoritative publicly available source for each model:

Model	Source for prior centres (x^{ctr})	Source for prior widths (σ_{ppm})
SP1	Hand-set defaults (no published global calib.)	Weakly informative
TP2	Hand-set ICBM-style defaults	Weakly informative
Yasso07	`y07par_gui.csv` (Tuomi et al.)	Literature CIs
Yasso15	`Yasso15.dat` posterior mean (FMI Ryassofortran)	`Yasso15.dat` posterior SD
Yasso20	`Yasso20_sample_parameters.rda` (FMI Ryassofortran)	`Yasso20.dat` posterior SD

The FMI repository is at https://github.com/YASSOmodel/Ryassofortran/tree/master/data. The `.dat` files contain 10,000 MCMC samples from the original calibrations, with row 1 being the MAP and rows 2–10,001 the posterior draws. For Yasso15/20, the MAP from `.rda` (or `y15par.csv`) and the posterior mean from `.dat` agree to four significant figures after column remapping (the original column ordering differs from HIKET's parameter-name ordering). The matching is verified in `Priors_model_matching.R`.

For Yasso20 the `.rda` file is preferred as the authoritative MAP because it is the FMI-validated point estimate used in the Ryassofortran package; `ParY20.dat` is retained as a cross-validation check.

For Yasso07 no posterior file is publicly available, so σ_{ppm} on its calibrated parameters comes from the confidence intervals reported in Tuomi et al. (2008–2010).

For SP1 and TP2 there is no global posterior to draw from. The defaults are set to plausible Yasso07-style starting values (e.g. β_1, β_2, γ copied from Yasso07 MAP, $\alpha = 0.09$ for SP1, ICBM-style α_A, α_H, p_H for TP2), and prior widths are set wide enough to let the Finnish data dominate.

4.2 Transfer-fraction priors

For all three Yasso models the prior standard deviation on every transfer fraction is held at $\sigma_{\text{ppm}} = 1.0$ in the unconstrained (stick-breaking) space. This is a deliberately weakly-informative choice: the Finnish data should drive the pool-flow structure, which is the primary structural-comparison axis of the intercomparison. The prior centres for the fractions are the published defaults (Yasso07: `y07par_gui.csv`; Yasso15: published defaults from `y15par.csv`; Yasso20: same Yasso15 defaults, since the previously-fixed zeros are now sampled and the Yasso15 MAP values are nonzero and from the same chemical-fractionation framework).

4.3 Climate-response and size-modifier priors

In contrast to the transfer fractions, the climate and size parameters receive *informative* priors, derived from the FMI posterior. This prevents two pathologies:

1. The γ / σ_{input} compensation loop. Both parameters scale total decomposition; without an informative prior on γ they trade off freely, producing a ridge in the posterior that destabilises MCMC mixing.
2. The $\delta_2 < 0$ NaN pathology. The Fortran size modifier returns NaN for $\delta_2 < 0$ at $d > 0.4$ cm; a log transform combined with an informative prior keeps δ_2 in the well-behaved region.

γ_H for Yasso15 is capped at $\sigma_{\text{ppm}} = 1.5$: the H-pool precipitation response is near-unidentifiable at Finnish precipitation levels ($\sim$ 600 mm), where $[1 - \exp(\gamma_H P/1000)] \approx 1$ for any $\gamma_H < -5$. The raw posterior SD ($\sim$ 3) reflects a flat likelihood surface, not genuine uncertainty.

4.4 Numerical values — Yasso07

Parameter	x^{ctr} (physical)	σ_{ppm}	Space
p_{ij} (12 fractions)	published defaults	1.0	unconstrained
β_1	0.0987	0.20	log
β_2	-0.00157	0.05	physical
γ	-1.272	0.30	physical
δ_1	-1.708	0.15	physical
δ_2	0.859	0.10	log
r $(= \lvert r \rvert)$	0.307	0.015	log
σ_{init}, σ_{input}	1.00	0.50	log

4.5 Numerical values — Yasso15

Parameter	x^{ctr}	σ_{ppm}	Space
p_{ij} (12 fr.)	published defaults	1.0	unconstr.
β_1	0.0906	0.0459	log
β_2	-2.15×10^{-4}	1.4×10^{-4}	physical
γ	-1.809	0.0702	physical
β_{N1}	0.0488	0.105	log
β_{N2}	-7.92×10^{-5}	8×10^{-5}	physical
γ_N	-1.173	0.155	physical
β_{H1}	0.0352	0.135	log
β_{H2}	-2.08×10^{-4}	1.6×10^{-4}	physical
γ_H	-12.54	1.50 (capped)	physical
δ_1	-0.439	0.328	physical
δ_2	1.267	0.286	log
r	0.257	0.055	log
σ_{init}, σ_{input}	1.00	0.50	log

4.6 Numerical values — Yasso20

Parameter	x^{ctr}	σ_{ppm}	Space
p_{ij} (12 fr.)	Yasso15 defaults	1.0	unconstr.
β_1	0.158	0.104	log
β_2	-0.002	5.4×10^{-4}	physical
γ	-1.440	0.146	physical
β_{N1}	0.170	0.0625	log
β_{N2}	-0.005	4.1×10^{-4}	physical
γ_N	-2.000	0.0353	physical

Parameter	x^{ctr}	σ_{ppm}	Space
β_{H1}	0.067	0.0912	log
β_{H2}	0.000	9×10^{-5}	physical
γ_H	-6.900	1.391	physical
δ_1	-2.550	0.435	physical
δ_2	1.240	0.211	log
r	0.250	0.0545	log
σ_{init}, σ_{input}	1.00	0.50	log

4.7 Numerical values — SP1 and TP2

Parameter	x^{ctr}	σ_{ppm}	Space
SP1			
α	0.09	0.50	log
β_1	0.095	0.20	log
β_2	-0.00014	0.05	physical
γ	-1.21	0.30	physical
TP2			
α_A	0.73	0.50	log
α_H	0.0015	0.50	log
p_H	0.028	1.00	logit
β_1	0.095	0.20	log
β_2	-0.00014	0.05	physical
γ	-1.21	0.30	physical
Both			
σ_{init}, σ_{input}	1.00	0.50	log

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5 Transient initialisation

5.1 Why transient initialisation

Finnish forest plots in the Biosoil network show, on average, a *positive* trend in mineral SOC from 1985 to 2006. Under a steady-state initial condition — where $C(1985) = C_{\text{ss}}$ given the litter inputs observed since 1985 — a model with first-order kinetics can only *increase* SOC if the litter input itself is increasing. The MUSTIKKA litter-input dataset does not show such an increase consistent in magnitude with the observed SOC trend. The observed SOC trajectory and the observed litter trajectory are, jointly, incompatible with a steady-state $C(1985)$ at the level implied by the modern litter inputs.

This is not a measurement artefact: it is a real signature that the forests in 1985 were not at steady state with respect to their then- recent productivity. They were either still recovering from earlier disturbance (less productive historically; SOC accumulating toward a higher steady state) or transitioning down from earlier higher productivity (more productive historically; SOC still decaying down). Either way, a calibration that enforces $C(1985) = J(\text{post-1985})/(\alpha\xi)$ has no degree of freedom available to fit the SOC trend, and either rejects all models or absorbs the discrepancy into implausible kinetic parameters.

5.2 The σ_{init} parameter

We introduce a single multiplicative scaling parameter, σ_{init}, applied to the litter inputs used to compute the initial state $C(1985)$:

$$C(1985) = C^*\big(\sigma_{\text{init}} \cdot \sigma_{\text{input}} \cdot J^{\text{ctr}}_{\text{litter}},\ \xi(\overline{\text{climate}})\big),$$

where $C^*(\cdot)$ is the model's analytical (or numerical) steady-state operator, $J^{\text{ctr}}_{\text{litter}}$ is the AWEN-fractionated litter vector over the first 20-year window (1985–2004), and $\xi(\overline{\text{climate}})$ is the climate modifier evaluated at the mean climate over the same window. σ_{input} continues to act on all subsequent litter inputs (1985 onward); σ_{init} acts *only* on the steady-state computation, so its physical effect is restricted to the initial pool state. The transient simulation then evolves $C(t)$ from $C(1985)$ under the actual time-varying litter and climate, with σ_{input} scaling those inputs but σ_{init} no longer entering.

The interpretation of the posterior of σ_{init} is:

- $\sigma_{\text{init}} \approx 1$: forests in 1985 were at steady state with respect to contemporary productivity;
- $\sigma_{\text{init}} > 1$: forests were *historically more productive* (old growth, pre-intensification), and 1985 SOC reflects that higher equilibrium;
- $\sigma_{\text{init}} < 1$: forests were *historically less productive* (post-disturbance recovery), and 1985 SOC is correspondingly lower.

The prior on σ_{init} is log-normal with centre 1 and unconstrained-space SD 0.5, giving a 95% physical-space prior interval of approximately $[0.37, 2.72]$. This is wide enough to accommodate plausible historical productivity ranges in Finnish forests while remaining centred at the no-change null.

5.3 Implementation across models

The pattern is uniform across all five models:

- `*_transient_init(model_params, lm, xi_mean)`: a function in each wrapper that takes the litter means `lm` (in particular, `lm$nwl_mean`, `lm$fwl_mean`, `lm$cwl_mean` for Yasso models, or `lm$J_total_mean` for SP1/TP2), multiplies them by $\sigma_{\text{input}} \cdot \sigma_{\text{init}}$, and passes the scaled litter to the model's existing steady-state routine.
- The engine's likelihood function then runs the transient simulation from this C_{init} over the full input time series with the unscaled-by-σ_{init} contemporary inputs.

The likelihood is also adjusted (see §6). In the *non-transient* calibration (the pre-existing pipeline, retained as a revert option in the engine), σ_{init} instead inflated the observation standard deviation σ_{obs} at the first observation per plot — effectively absorbing the initial-state uncertainty into the likelihood. With transient initialisation, σ_{init} has already entered the model dynamics; if it were also kept in the likelihood it would be double-counted. The engine flag `transient_init = TRUE` removes the patch.

5.4 Steady-state computation as the initial-state operator

We use the model's analytical (or Fortran-based) steady-state operator, not a Fortran pre-run, to compute $C(1985)$. For SP1, TP2 and the Yasso family, the steady state under constant inputs and constant ξ is mathematically well-defined and computable in closed form (SP1/TP2) or via a matrix solve (Yasso). A pre-run alternative would be to integrate the model from some earlier date (e.g. 1917, Finland's independence) at constant pre-1985 forcing. The analytical steady state is preferred because:

1. It is exact under the assumed forcing.
2. It avoids the choice of an arbitrary pre-run start date and an arbitrary pre-1985 productivity scenario.
3. For the fast AWEN pools (mean residence times 1–10 yr), any reasonable pre-run converges to the steady state within a decade, so the pre-run only retains information about H (MRT $\sim$ 600 yr); but H also barely changes over 68 yr of pre-run, so the approximation error is negligible.
4. The Fortran `yasso07_run` aggregates the 15-element per-cohort state into 5 summed pools in its output; reconstructing a 15-element C_{init} from this output causes numerical pathologies. The analytical steady-state routine returns the full 15-element state correctly.

5.5 Jensen's inequality correction

The steady-state climate modifier ξ is computed as $\xi(\text{cli}\bar{\text{m}}\text{ate})$, *not* as $\overline{\xi(\text{climate}_t)}$. Because ξ is convex in temperature (via $\exp(\beta_1 T + \beta_2 T^2)$) and concave in precipitation (via $1 - \exp(\gamma P/1000)$), Jensen's inequality gives $\overline{\xi(\text{climate}_t)} \geq \xi(\text{cli}\bar{\text{m}}\text{ate})$ for typical Finnish inter-annual variability. The bias is 5–15% in boreal conditions; large enough to materially affect C_{init}. Each wrapper provides a `compute_xi_mean_*` function that takes the steady-state climate window, averages each climate variable, and calls the main ξ function once with these scalar mean inputs.

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6 Likelihood

6.1 Form

Conditional on parameters, observations are modelled as

$$y_{p,t} \sim \mathcal{N}(\hat{C}_{p,t},\ \hat{C}_{p,t} \cdot \sigma_{\text{obs}}),$$

where $y_{p,t}$ is the observed SOC stock (t C ha^{-1}) at plot p in observation year t, and $\hat{C}_{p,t}$ is the predicted total SOC. This is a multiplicative-normal observation model: the standard deviation of the error scales with the prediction, so the relative observation error is constant across the SOC magnitude range.

6.2 Fixed σ_{obs}

σ_{obs} is *not* sampled. It is held at the empirical coefficient of variation of the SOC observations across all calibration plots and observation years, $\sigma_{\text{obs}} = \text{SD}(y)/\text{mean}(y)$. This is approximately 0.45–0.50 on the Finnish dataset.

This is a deliberate methodological choice. If σ_{obs} were sampled, a structurally inadequate model could inflate it to absorb its own residual variance, hiding the inadequacy. By fixing σ_{obs} at the empirical CV, the same observation-error budget is enforced across all five models, and structural differences have nowhere to hide except in the residuals — which is exactly where the intercomparison wants them to appear. The same value is used in all five calibrations.

6.3 Per-plot log-likelihood

For each plot p with observations at years $\{t_1, \dots, t_{n_p}\}$, the per-plot log-likelihood is

$$\log L_p = \sum_{k=1}^{n_p} \log \mathcal{N}(y_{p,t_k}\,;\,\hat{C}_{p,t_k},\ \hat{C}_{p,t_k} \cdot \sigma_{\text{obs}}).$$

The total log-likelihood is $\sum_p \log L_p$, plus the log-Jacobian of the parameter transforms. Plots are evaluated in parallel via `mclapply`.

6.4 Rejection logic

Three hard rejections (return $-\infty$) are applied, all required for arithmetic validity:

1. $\hat{C}_{p,t}$ is non-finite (NaN or $\pm\infty$) at any observation year — $\log \mathcal{N}$ undefined.
2. $\hat{C}_{p,t} \leq 0$ at any observation year — undefined in the multiplicative-error model.
3. The model run itself returned NULL (Fortran crash or wrapper-level `tryCatch` exception).

No physical-ceiling guard (e.g. "reject if predicted SOC > 1000 t C ha^{-1}") is applied. Such a ceiling would asymmetrically penalise models with recycling architectures that explore wider parameter regions during MCMC exploration. Physically implausible predictions yield very large negative log-likelihoods and are rejected by MCMC sampling naturally. The behaviour matches Viskari (2022).

The likelihood evaluates $\hat{C}_{p,t}$ only at observation years. Post-1985 trajectory values *between* observation years are computed (the simulation runs the full time series) but not entered into the likelihood. The full trajectory — including any structural divergence beyond the calibration window — is then examined in the predictive stage.

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7 MCMC sampling

The Differential Evolution with snooker update sampler (DEzs) is used, via the `BayesianTools` R package (Hartig et al. 2023). DEzs is a population-based sampler that proposes new states by taking weighted differences between current chain states, then accepts/rejects via the standard Metropolis ratio. It mixes well in moderate-dimensional problems (10–50 parameters) and handles correlated posteriors better than random-walk Metropolis.

7.1 Settings

Shared across all five models, defined in `calib_config.R`:

Setting	Value	Note
Chains	5	Sufficient for Gelman–Rubin $\hat{R}$
Iterations	50,000	Per chain
Burn-in	5,000	10% of iterations
Thinning	None	Full chain retained
Initial state	$x =$ best_x	All chains seeded at prior centre

DEzs initialises its internal population from small random perturbations around the seed point and then explores via differential-evolution proposals. `parallel = FALSE` is passed to `createBayesianSetup()` to suppress BayesianTools' internal parallelism; per-plot evaluation is already parallelised at the likelihood level.

7.2 Parallelisation

Two-level parallelism is available:

- **Per-plot, within-chain**: each likelihood evaluation runs the per-plot inner loop in parallel via `mclapply` with `CORES_PER_CHAIN` workers.
- **Across-chain (hybrid)**: chains can be launched in parallel on Linux clusters, with each chain then running its own per-plot parallelism.

The within-chain mode is mandatory on macOS (nested forking in mclapply is unsafe). On Linux HPC, hybrid mode (chains × cores) gives roughly linear speedup up to the available core count.

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8 Sanity checks and diagnostics

8.1 Pre-MCMC sanity check

Before launching MCMC, each calibration script performs a forward simulation at the prior centre ($x = $ best_x) for four randomly selected plots. Three conditions must hold:

1. All predicted SOC values are finite and positive.
2. The predicted-to-observed ratio lies in $[0.1, 10]$ at every observation year.
3. The likelihood `ll_fn(best_x)` is finite across all calibration plots.

A 2$×$2 diagnostic PNG is written (`{Model}_forward_at_defaults_{RUN_ID}.png`). If condition 1 or 3 fails, the calibration is aborted; condition 2 produces a warning but does not abort.

A complementary set of *input* diagnostics (`diagnostics_inputs.R`, run separately) confirms that the per-plot litter inputs are not so low that the implied mean residence time (MRT = mean SOC / mean litter) exceeds 100 yr, which would prevent the model from producing positive SOC at any parameter setting. Plots failing this test are excluded upstream via `calib_ready = FALSE`.

8.2 Convergence diagnostics

Computed on post-burnin samples in *unconstrained* (sampling) space:

Diagnostic	Threshold	Interpretation
$\hat{R}$ (univariate, per parameter)	< 1.05	Good convergence
$\hat{R}$ (multivariate)	< 1.05	Joint convergence
Effective sample size	> 200	Adequate independent draws
Acceptance rate	0.2–0.4	Reasonable DEzs mixing

$\hat{R} > 1.10$ is treated as poor convergence; the run is not used for downstream analysis. Per-chain divergence and post-warmup means are also tracked.

8.3 Diagnostic outputs

All written to `Calibration_real_data_transient/diagnostics/{Model}/`:

File	Content
`*_traces_*.png`	Trace plots, one panel per parameter
`*_marginals_*.png`	Prior vs posterior KDE (shared x-grid)
`*_rhat_barplot_*.png`	$\hat{R}$ per parameter, coloured by zone
`*_kl_divergence_*.png`	KL divergence posterior ‖ prior
`*_report_*.txt`	Tabular $\hat{R}$, ESS, quantile summaries
`*_forward_at_defaults_*.png`	Pre-MCMC sanity check (4 plots)

Prior and posterior KDE are evaluated on the same x-grid derived from the prior's 99th-percentile range. This is essential: tight priors otherwise appear artificially flat (compressed off-grid) when the x-axis is chosen from the posterior range, masking the actual prior–posterior displacement.

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9 Output files

Written to `Calibration_real_data_transient/runs/`:

File	Content	Used by
`{Model}_chains_{RUN_ID}.rds`	BayesianTools chain objects (unconstrained space)	Re-diagnostics, extension
`{Model}_posterior_{RUN_ID}.rds`	Post-burnin samples in physical space	Predictive scripts
`{Model}_metadata_{RUN_ID}.rds`	Run configuration, wallclock, $\hat{R}$, ESS	Reproducibility
`{Model}_inputs_{RUN_ID}.rds`	Per-plot input structures used by this calibration	Predictive scripts

The inputs RDS contains `climate_by_plot`, `inputs_by_plot`, `litter_means`, `obs_meta`, `plot_info`, `plots_real`, `sigma_obs_fixed`, `sigma_ppm`, and `STEADY_STATE_YEARS`. It provides exact provenance: the predictive script reads the same per-plot structures the calibration used, regardless of subsequent updates to `Data_work.R`.

10 Key design decisions

The following architectural choices are locked in across the pipeline:

1. **Transient initialisation via σ_{init}.** The 1985 SOC trajectory is incompatible with a steady-state initial condition under the observed litter inputs; σ_{init} gives the model freedom to start above or below the litter-implied steady state. Uniform across all five models.

2. **Multiplicative-normal likelihood.** Reverted from an earlier log-normal workaround that had been added when litter inputs were over-estimated by an order of magnitude; with correctly-scaled litter, multiplicative-normal is the appropriate model.

3. **Fixed σ_{obs}** at the empirical CV of the SOC observations, identical across all five models. Forces structural inadequacy into residuals rather than into the error term.

4. $\xi(\text{cli}\bar{\text{m}}\text{ate})$ **rather than** $\overline{\xi(\text{climate}_t)}$ for steady-state initialisation. Jensen's-inequality correction; 5–15% bias in boreal conditions otherwise. Implemented uniformly via `compute_xi_mean_*` functions in each wrapper.

5. **Steady-state climate window = 20 years.** 1985–2004 climate is used for the steady-state ξ, pre-dating the accelerated boreal warming of the 2010s.

6. **Decomposition rates fixed at published defaults** for the Yasso family $(\alpha_A, \alpha_W, \alpha_E, \alpha_N, p_H, \alpha_H)$. Keeps the intercomparison clean and reduces free-parameter count. SP1 and TP2 have no published global calibrations, so all their parameters are free.

7. **Log transforms on δ_2 and r.** Enforces $\delta_2 > 0$ to avoid Fortran NaN at $d > 0.4$ cm and eliminates sign-redundancy in $|r|$.

8. **Informative priors on climate and size parameters** (Yasso15/20 from FMI posterior SDs, Yasso07 from literature CIs). Prevents the $\gamma/\sigma_{\text{input}}$ compensation loop and the δ_2 NaN pathology.

9. **Weakly informative priors on transfer fractions** ($\sigma_{\text{ppm}} = 1$ in unconstrained space). Finnish data drives pool-flow structure — the primary structural-comparison axis.

10. **All 12 transfer fractions free in all three Yasso models.** The Viskari (2022) structural zeros and derived fractions in the original Yasso20 are *not* imposed; doing so would give Yasso20 fewer degrees of freedom than Yasso07/15 and make the structural intercomparison uninterpretable.

11. **Stick-breaking parameterisation** with budget $B = 1 - p_H$ for transfer fractions. Exact simplex constraint, well-conditioned Jacobian, exchangeable across donor columns.

12. **Three SOC observation years** (1985, 2006, 2024). The Komeetta campaign (2024) provides a third calibration target; the 2006–2024 quasi-stability constrains long-term decomposition rates independently of the 1985–2006 accumulation dynamics.

13. **No physical-ceiling rejection** in the likelihood. Implausible predictions are rejected via the log-likelihood, not via a hard cap. Matches Viskari (2022).

14. **SP1 and TP2 share the Yasso07 ξ formula.** Climate-response comparison across the complexity ladder is not confounded by approximation-scheme differences.
15. **ICBM topology for TP2**: all litter enters A; H receives only transferred mass. Eliminates a weakly-constrained input-partitioning parameter and is standard in Nordic forestry SOC models.
16. **Global σ_{input}.** A single multiplicative litter scaling per plot. Per-species or per-region stratification is a possible extension but is not used in the current pipeline.

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11 Software

Component	Details
R	$\geq$ 4.3
`BayesianTools`	Hartig et al. (2023); DEzs sampler
`coda`	Gelman–Rubin $\hat{R}$, ESS
`compiler`	`cmpfun()` byte-compilation of likelihood
`parallel`	`mclapply()` per-plot parallelism
Fortran 90 cores	`R CMD SHLIB`; `.Fortran()` interface (Yasso family)
SP1, TP2	Pure R; no Fortran compilation

Random seed: `set.seed(2025)` before chain initialisation, consistent across all models. Test plot subsampling (when `N_PLOTS_TEST` is set): `set.seed(42)`.

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12 References

- Andrén, O. & Kätterer, T. (1997). ICBM: The introductory carbon balance model for exploration of soil carbon balances. *Ecological Applications* 7, 1226–1236.
- Hartig, F. et al. (2023). BayesianTools: General-purpose MCMC and SMC samplers and tools for Bayesian statistics. R package.
- Tuomi, M. et al. (2008–2011). Yasso07 calibration and the leaf litter decomposition model. *Ecological Modelling* and *Global Change Biology*.
- Viskari, T. et al. (2022). Calibrating the soil organic carbon model Yasso20 with multiple Finnish datasets. *Geoscientific Model Development* 15, 1735–1752.